

# Byungsoo Oh

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## RESEARCH INTERESTS

Systems for ML, Hardware-Software Co-Design, Networking for AI Infrastructure

## EDUCATION

### Cornell University

Ph.D. in Computer Science

Advisor: Prof. Rachee Singh

Ithaca, NY, USA

Aug 2024 – Present

### Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Computer Science

Daejeon, South Korea

Feb 2020

### Sogang University

B.S. in Computer Science and Engineering

Graduated with honors, *Summa Cum Laude*

Seoul, South Korea

Feb 2018

## PROFESSIONAL EXPERIENCE

### Microsoft Research

Research Intern

Redmond, WA, USA

May 2025 – Aug 2025

### Samsung Research

Research Engineer

Seoul, South Korea

Feb 2020 – Jun 2024

## PUBLICATIONS

- Eric Ding, **Byungsoo Oh**, Bhaskar Kataria, Kaiwen Guo, Jelena Gvero, Abhishek Vijaya Kumar, Arjun Devraj, Lindsey Bowen, Atharv Sonwane, Emaad Manzoor, Rachee Singh, "CCL-Bench 1.0: A Trace-Based Benchmark for LLM Infrastructure", arXiv, 2026
- Byungsoo Oh**, Rachee Singh, "Eliminating Hidden Serialization in Multi-Node Megakernel Communication", arXiv, 2026
- Abhishek Vijaya Kumar\*, Bhaskar Kataria\*, **Byungsoo Oh**, Emaad Manzoor, Rachee Singh, "TVCache: A Tool-Value Cache for Post-Training LLM Agents", International Conference on Machine Learning (ICML), Seoul, South Korea, 2026 (\* denotes co-first authors)
- Osayamen Jonathan Aimuyo, **Byungsoo Oh**, Rachee Singh, "FlashMoE: Fast Distributed MoE in a Single Kernel", Conference on Neural Information Processing Systems (NeurIPS), San Diego, CA, USA, 2025
- Taegeon Um\*, **Byungsoo Oh**\*, Minyoung Kang\*, Woo-Yeon Lee, Goeun Kim, Dongseob Kim, Youngtaek Kim, Mohd Muzzammil, Myeongjae Jeon, "Metis: Fast Automatic Distributed Training on Heterogeneous GPUs", USENIX Annual Technical Conference (USENIX ATC), Santa Clara, CA, USA, 2024 (\* denotes co-first authors)
- Taegeon Um, **Byungsoo Oh**, Byeongchan Seo, Minhyeok Kweun, Goeun Kim, Woo-Yeon Lee, "FastFlow: Accelerating Deep Learning Model Training with Smart Offloading of Input Data Pipeline", International Conference on Very Large Data Bases (VLDB), Vancouver, Canada, 2023
- Minhyeok Kweun, Goeun Kim, **Byungsoo Oh**, Seongho Jung, Taegeon Um, Woo-Yeon Lee, "PokéMem: Taming Wild Memory Consumers in Apache Spark", IEEE International Parallel and Distributed Processing Symposium (IPDPS), Lyon, France, 2022
- Seungju Cho, Tae Joon Jun, **Byungsoo Oh**, Daeyoung Kim, "DAPAS: Denoising Autoencoder to Prevent Adversarial attack in Semantic Segmentation", International Joint Conference on Neural Networks (IJCNN), Glasgow, UK, 2020
- Byungsoo Oh**, Daeyoung Kim, "Serverless-Enabled Permissioned Blockchain for Elastic Transaction Processing", ACM/IFIP International Middleware Conference (Middleware), Poster Paper, Davis, CA, USA, 2019
- Byungsoo Oh**, Tae Joon Jun, Wondeuk Yoon, Yunho Lee, Sangtae Kim, Daeyoung Kim, "Enhancing Trust of Supply Chain Using Blockchain Platform with Robust Data Model and Verification Mechanisms", IEEE International Conference on Systems, Man, and Cybernetics (SMC), Bari, Italy, 2019

## PATENTS

- Minyoung Kang, **Byungsoo Oh**, Taegeon Um, "Method and System for Elastic Knowledge Distillation with Adaptive Coordination", US Patent, US20250068943A1, Published: Feb 27, 2025
- Taegeon Um, Minhyeok Kweun, **Byungsoo Oh**, "Smart Offloading for AI Input Data Pipeline Acceleration", US Patent, US20240135189A1, Published: Apr 25, 2024
- Minyoung Kang, **Byungsoo Oh**, Taegeon Um, "Device Placement Strategies for Optimizing 3D Parallelism in Non-Uniform Topology Environments", US Patent, Pending, 2023
- Daeyoung Kim, **Byungsoo Oh**, "Method and System for Enhancing Trust of Supply Chain Using Blockchain Platform with Robust Data Model and Verification Mechanisms", Korean Patent, No. 10-2620822-0000, Issued: Dec 2023

## HONORS AND AWARDS

- **LinkedIn Fellowship** 2025–2026  
Cornell Bowers CIS-LinkedIn grant for academic year of 2025-2026
- **USENIX ATC 2024 Student Grant** 2024  
Travel grant awarded to attend USENIX ATC 2024 (co-located with OSDI 2024) in Santa Clara
- **National Full Scholarship**, Korea Ministry of Science and ICT 2018–2020
- **Award for Top 1% Students in College of Engineering (Dean's List)**, Sogang University 2017  
2 semesters (Spring 2017, Fall 2017)
- **Academic Excellence Scholarship**, Sogang University 2013–2017  
6 semesters (Spring 2013, Fall 2015, Spring 2016, Fall 2016, Spring 2017, Fall 2017)

## ACADEMIC SERVICE

- **Artifact Evaluation Committee:** SOSP 2025

## TEACHING EXPERIENCE

- TA, Systems for Large-Scale Machine Learning (CS5470), Cornell University Fall 2026
- TA, Introduction to Computer Networks (CS4450/5456), Cornell University Fall 2024
- TA, Introduction to System Programming (CS230), KAIST Spring 2019, Spring 2018
- TA, Embedded Operating Systems (CS632), KAIST Fall 2018

## TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Go, JavaScript, Java, Scala, Markdown,  $\text{\LaTeX}$
- **ML Systems:** PyTorch, TensorFlow, DeepSpeed, Megatron-LM, Alpa, vLLM, verl, CUDA, NSight Systems & Compute
- **Infra & Tools:** Docker, Kubernetes, gRPC, Apache Airflow, Apache Spark, Apache Hadoop

## OPEN SOURCE CONTRIBUTIONS

- **DeepSpeed.** Bug Fix [\[issue\]](#) [\[code\]](#)
- **TensorFlow.** Documentation improvement for `tf.data` service [\[code\]](#)
- **Apache Spark.** Bug Fix [\[code\]](#), Benchmark [\[code\]](#)

## LANGUAGES

Korean (*native*), English (*fluent*)